Nikolaos Pappas

Principal Scientist in Multimodal AI, Safety, Efficiency, and Continual Learning
nik0spapp@gmail.com | http://nik0spapp.github.io | 7100+ citations, h-index: 24, i10-index: 27

Summary

Principal scientist with a decade of experience focusing on multimodal modeling, safety alignment, efficient architectures, and continual learning through no-regret update strategies. My work in the tech industry has led to 6 product launches, 7 filed patents, and 18 publications in top-tier NLP and deep learning venues. I combine deep research expertise with real-world impact, driving innovations in cross-modal moderation, robust adaptation, and scalable deployment of foundation models.

Selected Contributions

Multimodal AI & Safety	*Defined the vision for multimodal safety in Guardrails, deploying the first enterprise-scale cross-modal moderation system to improve safety precision. Orchestrated an adapter-based customization framework for open-weight models (training, inference, eval) that achieves superior performance over prompting and joint or independent finetuning. Developed an agentic prompt optimization and benchmarking workflow reducing model update time by up to 94%. Initiated RAI/SDG collaborations for adversarial attack defense design and policy-based data generation; expanded safety coverage to code modality.*
Robust & Efficient Adaptation	*Drove core research for Amazon's conversational AI and Titan LLMs, focusing on continual learning and model compression. Developed stable, no-regret fine-tuning strategies that reduced catastrophic forgetting, preserved finetuned accuracy, cut data requirements, and minimized regression rates in production. Techniques deployed across multiple customer-facing systems.*
Scalable Attention	*Initiated and directed a research trajectory on scalable attention that led to linear-time architectures (e.g., Linear Attention, RFA). Defined the problem desiderata and proposed the use of a kernel-based formulation that underpinned multiple models. Contributions led to multiple publications and inspired extensions in memory-bounded retrieval, document-level translation, and long-context pretraining.*
Business Impact	*Research has powered 6 product launches, 7 filed patents, and 18 publications (including Best Theme Paper at EMNLP 2025). Mentored 15+ applied scientists, conducted 23+ interviews, and contributed to 6 code libraries with wide production adoption.*

Employment history

3.2026-Present	**Oracle Cloud Infrastructure**, *Lead Principal Applied Scientist*, GenAI org, Seattle, USA.
2.2022-2.2026	**Amazon Web Services**, *Senior Applied Scientist*, AWS AI labs/DS3, Seattle, USA.
7.2021–2.2022	**Amazon Web Services**, *Applied Scientist*, AWS AI labs, Seattle, USA.
9.2019-7.2021	**University of Washington**, *Postdoctoral Researcher*, Noah's ARK NLP group, Seattle, USA.
4.2016-8.2019	**Idiap Research Institute**, *Postdoctoral Researcher*, NLP group, Martigny, Switzerland.
2.2012-3.2016	**EPFL & Idiap Research Institute**, *Research Assistant*, NLP group, Martigny, Switzerland.
7.2014-12.2014	**Columbia University**, *Visiting Scholar*, DVMM lab, New York, USA.
5.2009-10.2011	**I-Sieve Technologies**, *Software Engineer*, Athens, Greece.
11.2007-11.2011	**Endysis Group**, *Software Engineer*, Athens, Greece.
3.2007-7.2007	**Optimor Labs**, *Software Developer*, Oxford, England.

Education

2016	**Ecole Polytechnique Fédérale de Lausanne (EPFL)**, *Lausanne, Switzerland.*
	Ph.D. Electrical Engineering
Thesis	Learning explainable user sentiment and preferences for information filtering
Advisors	Andrei Popescu-Belis, Hervé Bourlard
2011	**University of the Aegean**, Samos, Greece.

M.Sc. Information Management

2009 **University of the Aegean**, Samos, Greece.
Dipl. Eng. Information and Communication Systems

Awards & Honors

2018 **Two-year postdoc fellowship to be hosted at the University of Washington**.
Swiss National Science Foundation

2017 **Best paper award "Multilingual Hierarchical Attention Networks for Document Classification"**.
8th International Joint Conference on Natural Language Processing, 2017

2016 **Best multimodal paper award "Multilingual Visual Sentiment Concept Matching"**.
ACM International Conference on Multimedia Retrieval, 2016

2015 **Best paper award "Adaptive Sentiment-aware One-Class Collaborative Filtering"**.
Idiap Research Institute

2013 **Ranked first on video hyperlinking task with Idiap's NLP group**.
MediaEval, ACM MM Conference 2013

2010 **Award for top graduates of Greek Polytechnic Schools (2009 – 2010)**.
TEE (Technical Chamber of Greece)

2006 **Scholarship for the third academic year (ranked 1st)**.
IKY (Greek State Scholarships Foundation)

2004 **Scholarship for the first academic year (ranked 2nd)**.
IKY (Greek State Scholarships Foundation)

Research funding

2019 **Swiss National Science Foundation**, *2 years*, Applicant, Switzerland.
UNISON, `http://p3.snf.ch/project-183911`.
Swiss National Science Foundation, *6 months*, Member, Switzerland.
INTREPID, `http://p3.snf.ch/Project-180320`.

2018 **Intelligence Advanced Research Projects Activity (IARPA)**, *3 months*, Member, USA.
MATERIAL, `http://summa-project.eu/`.

2016 **European Union (Horizon 2020)**, *3 years*, Member, Switzerland.
SUMMA, `http://summa-project.eu/`.

2015 **TheArk Foundation**, *6 months*, Member, Switzerland.
HYBRID, `https://www.theark.ch/en/`.

2014 **Swiss National Science Foundation**, *1 month*, Member, Switzerland.
AROLES, `http://p3.snf.ch/Project-144627`.

2012 **European Union (FP7-ICT)**, *4 years*, Member, Switzerland.
InEvent, `https://www.inevent-project.eu/`.

Teaching

Ecole Polytechnique Fédérale de Lausanne (EPFL)

2016 EE-724 – Human Language Technology: Applications to Information Access
4 guest lectures.

2012 EE-724 – Human Language Technology: Applications to Information Access
1 guest lecture.

Mentoring

University of Washington

2020 Daniel Zhu – UW CSE BS student, 01/2021 – 07/2021.

Stephen Wu – UW CSE BS student, 02/2019 – 07/2021.

Sean Hung – UW CSE BS student, 10/2020 – 12/2020.

2019 Xuhui Zhou – UW CLMS MS student, 10/2019 – 06/2020.

Ivan Montero – UW CSE BS student, 10/2019 – 07/2021.

Ecole Polytechnique Fédérale de Lausanne (EPFL)

2017 Khalil Mrini – EPFL CS MS student, 03/2017 – 09/2017.

Jonathan Pilault – Idiap intern, 03/2017 – 09/2017.

Academic Service

NLP **EMNLP**, *Conference on Empirical Methods on Natural Language Processing*, 2017-2026.

ACL, *Annual Meeting of the Association for Computational Linguistics*, 2017, 2019-2026.

NAACL-HLT, *Annual Conference of the North American Chapter of the Association for Computational Linguistics*, 2018-2026.

AACL-IJCNLP, *Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 9th International Joint Conference on Natural Language Processing*, 2020.

ACL RR, *Association for Computational Linguistics Rolling Review*, 2021-Present.

TACL, *Transactions of the Association for Computational Linguistics*, 2021.

COLING, *International Conference on Computational Linguistics*, 2016, 2020.

LREC, *Language Resources and Evaluation Conference*, 2018, 2020.

ML/Other **ICML**, *International Conference on Machine Learning*, 2020-2026.

ICLR, *International Conference on Learning Representations*, 2021-2026.

NeurIPS, *Conference on Neural Information Processing Systems*, 2021-2026.

AAAI, *Conference on Artificial Intelligence*, 2020-2022.

IJCAI, *International Joint Conference on Artificial Intelligence*, 2021.

ICMR, *International Conference on Multimedia Retrieval*, 2017-2019.

MM, *ACM Multimedia*, 2017-2018.

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AC/SAC duties, *EACL 2023, EACL 2024, EACL 2026, NAACL-HLT 2024, ACL 2024, ACL 2026, ICML 2027.*

Talks & Lectures

2021 **Large-context and efficient models of language**
Pindrop, Online.

2020 **Grounded Compositional Outputs for Adaptive Language Modeling**
EMNLP conference, Online.

2019 **Deep Residual Output Layers for Neural Language Generation**
ICML Conference, Long Beach, CA, USA.

2018 **Basics and Modern Methods for Natural Language Processing**
UnternehmerTUM, Munich, Germany.

2017 **Multilingual Hierarchical Attention Networks for Document Classification**
IJCNLP conference, Tapei, Taiwan.

Labeling Text in Several Languages with Multilingual Hierarchical Attention Networks, Swisstext, Zurich, Switzerland.

2016 **Deep Learning for NLP: Multilingual Word Sequence Modeling**
EE-724 EPFL, Lausanne, Switzerland.

Human versus Machine Attention in Document Classification: A Dataset with Crowdsourced Annotation, EMNLP conference, Austin, Texas.

Deep Learning for NLP: Word Representation Learning

EE-724 EPFL, Lausanne, Switzerland.

Explicit Document Modeling using Multiple-Instance Learning

Data Science group at Swisscom, Lausanne, Switzerland.

Labeling Text in Several Languages with Multilingual Hierarchical Attention Networks, Swisstext, Zurich, Switzerland.

2015 **Weighted Multiple Instance Learning for Text Regression Tasks**

Swiss Machine Learning Days, Lausanne Switzerland.

2013 **Learning Aspects and Ratings with Multiple-Instance Regression from Text**

Swiss Machine Learning Days, Lausanne Switzerland.

Combining Content with User Preferences for TED Lecture Recommendation

CBMI conference, Veszprém, Hungary.

2012 **Intoduction to Recommender Systems**

EE-724 EPFL, Lausanne, Switzerland.

Journal publications

Resulting from Postdoctoral time

2019 [6] Nikolaos Pappas, James Henderson, "GILE: A Generalized Input-Label Embedding for Text Classification", In *Transactions of the Association for Computational Linguistics* (TACL), 2019. URL `https://arxiv.org/pdf/1806.06219.pdf`.

2018 [5] Xiao Pu, Nikolaos Pappas, James Henderson, Andrei Popescu-Belis, "Integrating Weakly Supervised Word Disambiguation into Neural Machine Translation", In *Transactions of the Association for Computational Linguistics* (TACL), 2018. URL `https://arxiv.org/pdf/1810.02614.pdf`.

2017 [4] Nikolaos Pappas, Andrei Popescu-Belis, "Explicit Document Modeling through Weighted Multiple-Instance Learning", *Journal of Artificial Intelligence Research* (JAIR), pages 591–626, 2017. URL `https://jair.org/index.php/jair/article/view/11051/26228`.

Resulting from PhD time

2017 [3] Nikolaos Pappas, Mercan Topkara, Miriam Redi, Hongyi Liu, Brendan Jou, Tao Chen, Shih-Fu Chang, "Multilingual Visual Sentiment Concept Clustering and Analysis", *International Journal of Multimedia Information Retrieval* (IJMIR), pages 1–20, 2017. URL `https://link.springer.com/article/10.1007/s13735-017-0120-4`.

2016 [2] Nikolaos Pappas, Andrei Popescu-Belis, "Adaptive Sentiment-Aware One-Class Collaborative Filtering", *Expert Systems with Applications* (ESWA), 43(1): 23-41, 2016. URL `https://www.sciencedirect.com/science/article/pii/S0957417415005916`.

2015 [1] Nikolaos Pappas, Andrei Popescu-Belis, "Combining Content with User Preferences for Non-Fiction Multimedia Recommendation: A Study on TED Lectures", *Multimedia Tools and Applications, Special Issue on Content-Based Multimedia Indexing* (MTAP), 74(4): 1175-1197, 2015. URL `https://link.springer.com/article/10.1007/s11042-013-1840-y`.

Conference and workshop publications

2026 [42] Duygu Nur Yaldiz, Evangelia Spiliopoulou, Zheng Qi, Siddharth Varia, Srikanth Doss, Nikolaos Pappas, "Balancing Classification and Calibration Performance in Decision-Making LLMs via Calibration Aware Reinforcement Learning", In *Findings of the Association for Computational Linguistics: ACL*, 2026. URL `https://aclanthology.org/2026.findings-acl.610.pdf`.

[41] Sadat Shahriar, Zheng Qi, Nikolaos Pappas, Srikanth Doss, Kishaloy Halder, Monica Sunkara, Manuel Mager, Yassine Benajiba, "MEAV: Model Editing with Alignment Vectors for inference time LLM alignment in single and multidomain preference spectrum", In *Findings of the Association for Computational Linguistics: ACL*, 2026. URL https://aclanthology.org/2026.findings-acl.2035.pdf.

[40] Rheeya Uppaal, Phu Mon Htut, Min Bai, Nikolaos Pappas, Zheng Qi, Sandesh Swamy, "Journey Before Destination: On the importance of Visual Faithfulness in Slow Thinking", In *European Chapter of the Association for Computational Linguistics* (EACL), 2026. URL https://arxiv.org/pdf/2512.12218.

2025 [39] Yu Feng, Phu Mon Htut, Zheng Qi, Wei Xiao, Manuel Mager, Nikolaos Pappas, Kishaloy Halder, Yang Li, Yassine Benajiba, Dan Roth, "Rethinking LLM uncertainty: A multi-agent approach to estimating black-box model uncertainty", In *Conference on Empirical Methods in Natural Language Processing* (EMNLP Findings), 2025. URL https://aclanthology.org/anthology-files/pdf/findings/2025.findings-emnlp.660.pdf.

[38] Qin Liu, Chao Shang, Ling Liu, Nikolaos Pappas, Jie Ma, Neha Anna John, Srikanth Doss, Lluis Marquez, Miguel Ballesteros, Yassine Benajiba, "Unraveling and mitigating safety alignment degradation of vision-language models", In *Annual Meeting of the Association for Computational Linguistics* (ACL Findings), 2025. URL https://arxiv.org/pdf/2410.09047?.

[37] James Y Huang, Sailik Sengupta, Daniele Bonadiman, Yi-an Lai, Arshit Gupta, Nikolaos Pappas, Saab Mansour, Katrin Kirchoff, Dan Roth, "DEAL: Decoding-time alignment for large language models", In *Annual Meeting of the Association for Computational Linguistics* (ACL), 2025. URL https://arxiv.org/pdf/2402.06147.

[36] Siyi Liu, Kishaloy Halder, Zheng Qi, Wei Xiao, Nikolaos Pappas, Phu Mon Htut, Neha Anna John, Yassine Benajiba, Dan Roth, "Towards Long Context Hallucination Detection", In *Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics* (NAACL Findings), 2025. URL https://arxiv.org/pdf/2504.19457?.

2024 [35] Devang Kulshreshtha, Saket Dingliwal, Brady Houston, Nikolaos Pappas, Srikanth Ronanki, "Sequential Editing for Lifelong Training of Speech Recognition Models", In *Interspeech Conference* (INTERSPEECH), 2024. URL https://arxiv.org/pdf/2406.17935?.

[34] Dhananjay Ram, Aditya Rawal, Momchil Hardalov, Nikolaos Pappas, Sheng Zha, "DEM: Distribution Edited Model for Training with Mixed Data Distributions", In *Conference on Empirical Methods in Natural Language Processing* (EMNLP), 2024. URL https://aclanthology.org/2024.emnlp-main.1074v2.pdf.

[33] Bryan Li, Tamer Alkhouli, Daniele Bonadiman, Nikolaos Pappas, Saab Mansour, "Eliciting Better Multilingual Structured Reasoning from LLMs through Code", In *Annual Meeting of the Association for Computational Linguistics* (ACL), 2024. URL https://arxiv.org/pdf/2403.02567.

[32] Hossein Aboutalebi, Hwanjun Song, Yusheng Xie, Arshit Gupta, Justin Sun, Hang Su, Igor Shalyminov, Nikolaos Pappas, Siffi Singh, Saab Mansour, "MAGID: An Automated Pipeline for Generating Synthetic Multi-modal Datasets", In *Annual Conference of the North American Chapter of the Association for Computational Linguistics* (NAACL), 2024. URL https://arxiv.org/pdf/2403.03194.

2023 [31] Shufan Wang, Sebastien Jean, Sailik Sengupta, James Gung, Nikolaos Pappas, Yi Zhang, "Measuring and Mitigating Constraint Violations of In-Context Learning for Utterance-to-API Semantic Parsing", In *Conference on Empirical Methods in Natural Language Processing.* (EMNLP Findings), 2023. URL https://arxiv.org/pdf/2305.14827.pdf.

[30] Mujeen Sung, James Gung, Elman Mansimov, Nikolaos Pappas, Raphael Shu, Salvatore Romeo, Yi Zhang, Vittorio Castelli, "Pre-training Intent-Aware Encoders for Zero-and Few-Shot Intent Classification", In *Conference on Empirical Methods in Natural Language Processing.* (EMNLP), 2023. URL https://arxiv.org/pdf/2305.14827.pdf.

[29] Zhanpeng Zeng, Cole Hawkins, Mingyi Hong, Aston Zhang, Nikolaos Pappas, Vikas Singh, Shuai Zheng, "Vcc: Scaling Transformers to 128K Tokens or More by Prioritizing Important Tokens", In *Conference on Neural Information Processing Systems.* (NeurIPS), 2023. URL https://arxiv.org/pdf/2305.04241.pdf.

[28] Shamik Roy, Raphael Shu, Nikolaos Pappas, Elman Mansimov, Yi Zhang, Saab Mansour, Dan Roth, "Conversation style transfer using few-shot learning", In *Joint Conference on Natural Language Processing and Asia-Pacific Chapter of the Association for Computational Linguistics.* (IJCNLP-AACL), 2023. URL https://arxiv.org/pdf/2302.08362.pdf.

[27] Raphael Schumann, Elman Mansimov, Yi-An Lai, Nikolaos Pappas, Xibin Gao, Yi Zhang, "Backward compatibility during data updates by weight interpolation", In *Conference of the European Chapter of the Association for Computational Linguistics.* (EACL Findings), 2023. URL https://arxiv.org/pdf/2301.10546.pdf.

Resulting from Postdoctoral time

2022 [26] Zhaofeng Wu, Hao Peng, Nikolaos Pappas, Noah A. Smith, "Modeling Context With Linear Attention for Scalable Document-Level Translation", In *In Findings of the Conference on Empirical Methods in Natural Language Processing* (EMNLP Findings), 2022. URL https://arxiv.org/pdf/2210.08431.pdf.

[25] Hao Peng, Jungo Kasai, Nikolaos Pappas, Dani Yogatama, Zhaofeng Wu, Lingpeng Kong, Roy Schwartz, Noah A. Smith, "ABC: Attention with Bounded-memory Control", In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics* (ACL), 2022. URL https://arxiv.org/pdf/2110.02488.pdf.

2021 [24] Ivan Montero, Nikolaos Pappas, Noah A. Smith, "Sentence Bottleneck Autoencoders from Pretrained Transformer Models", In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (EMNLP), 2021. URL https://aclanthology.org/2021.emnlp-main.137.pdf.

[23] Jungo Kasai, Hao Peng, Yizhe Zhang, Dani Yogatama, Gabriel Ilharco, Nikolaos Pappas, Yi Mao, Weizhu Chen, Noah A. Smith, "Finetuning Pretrained Transformers into RNNs", In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (EMNLP), 2021. URL https://aclanthology.org/2021.emnlp-main.830.pdf.

[22] Hao Peng, Nikolaos Pappas, Dani Yogatama, Roy Schwartz, Noah Smith, Lingpeng Kong, "Random Feature Attention", In *Proceedings of the International Conference on Learning Representations* (ICLR), 2021. URL https://openreview.net/pdf?id=QtTKTdVrFBB.

[21] Jungo Kasai, Nikolaos Pappas, Hao Peng, James Cross, Noah Smith, "Deep Encoder, Shallow Decoder: Reevaluating Non-autoregressive Machine Translation", In *Proceedings of the International Conference on Learning Representations* (ICLR), 2021. URL https://openreview.net/pdf?id=KpfasTaLUpq.

2020 [20] Nikolaos Pappas, Phoebe Mulcaire, Noah A. Smith "Grounded Compositional Outputs for Adaptive Language Modeling", In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (EMNLP), 2020. URL https://aclanthology.org/2020.emnlp-main.96.pdf.

[19] Xuhui Zhou, Nikolaos Pappas, Noah A. Smith "Multilevel Text Alignment with Cross-Document attention", In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (EMNLP), 2020. URL https://aclanthology.org/2020.emnlp-main.407.pdf.

[18] Florian Mai, Nikolaos Pappas, Ivan Montero, Noah A. Smith, James Henderson "Plug and Play Autoencoders for Conditional Text Generation", In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (EMNLP), 2020. URL https://aclanthology.org/2020.emnlp-main.491.pdf.

[17] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, Francois Fleuret, "Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention", In *Proceedings of the 37th International Conference on Machine Learning* (ICML), Vienna, Austria, 2020. URL http://proceedings.mlr.press/v119/katharopoulos20a/katharopoulos20a.pdf.

2019 [16] Nikolaos Pappas, James Henderson, "Deep Residual Output Layers for Neural Language Generation", In *Proceedings of the 36th International Conference on Machine Learning* (ICML), Long Beach, California, 2019. URL `http://publications.idiap.ch/downloads/papers/2019/Pappas_ICML_2019.pdf`.

2018 [15] Lesly Miculicich Werlen, Dhananjay Ram, Nikolaos Pappas, James Henderson, "Document-Level Neural Machine Translation with Hierarchical Attention Networks", In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing* (EMNLP), Brussels, Belgium, 2018. URL `http://publications.idiap.ch/downloads/papers/2018/Miculicich_EMNLP_2018.pdf`.

[14] Nikolaos Pappas, Lesly Miculicich Werlen, James Henderson, "Beyond Weight Tying: Learning Joint Input-Output Embeddings for Neural Machine Translation", In *Proceedings of the 2018 Third Conference on Machine Translation* (WMT), Brussels, Belgium, 2018. URL `http://publications.idiap.ch/downloads/papers/2018/Pappas_WMT_2018.pdf`.

[13] Lesly Miculicich Werlen, Nikolaos Pappas, Dhananjay Ram, Andrei Popescu-Belis, "Self-attentive Residual Decoder for Neural Machine Translation", In *Proceedings of the 16th Annual Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (NAACL), pages 1366–1379, New Orleans, USA, 2018. URL `http://aclweb.org/anthology/N18-1124`.

2017 [12] Nikolaos Pappas, Andrei Popescu-Belis, "Multilingual Hierarchical Attention Networks for Document Classification", In *8th International Joint Conference on Natural Language Processing* (IJCNLP), pages 1015–1025, Tapei, Taiwan, 2017. URL `http://aclweb.org/anthology/I17-1102`.

[11] Xiao Pu, Nikolaos Pappas, Andrei Popescu-Belis, "Sense-Aware Statistical Machine Translation using Adaptive Context-Dependent Clustering", In *Proceedings of Second Conference on Machine Translation* (WMT), pages 1–10, Copenhagen, Denmark, 2017. URL `http://www.statmt.org/wmt17/pdf/WMT01.pdf`.

2016 [10] Nikolaos Pappas, Andrei Popescu-Belis, "Human versus Machine Attention in Document Classification: A Dataset with Crowdsourced Annotations", In *Proceedings of 2016 Empirical Methods for Natural Language Processing SocialNLP Workshop* (SocialNLP), pages 94–100, Austin, USA, 2016. URL `http://www.aclweb.org/anthology/W16-6213`.

Resulting from PhD time

2016 [9] Nikolaos Pappas, Mercan Topkara, Miriam Redi, Brendan Jou, Tao Chen, Hongyi Liu, Shih-Fu Chang , "Multilingual Visual Sentiment Concept Matching", In *Proceedings of the ACM International Conference on Multimedia Retrieval* (ICMR), pages 151–158, New York, NY, USA, 2016. URL `http://delivery.acm.org/10.1145/2920000/2912016/p151-pappas.pdf`.

[8] Hongyi Liu, Brendan Jou, Tao Chen, Mercan Topkara, Nikolaos Pappas, Miriam Redi, Shih-Fu Chang , "Complura: Exploring and Leveraging a Large-scale Multilingual Visual Sentiment Ontology", In *Proceedings of the 2016 ACM on International Conference on Multimedia Retrieval* (ICMR), pages 417–420, New York, NY, USA, 2016. URL `https://dl.acm.org/citation.cfm?id=2911996.2912030`.

2015 [7] Brendan Jou*, Tao Chen*, Nikolaos Pappas*, Miriam Redi*, Mercan Topkara* and Shih-Fu Chang, "Visual Affect Around the World: A Large-scale Multilingual Visual Sentiment Ontology", In *Proceedings of ACM International Conference on Multimedia* (ACMMM), Brisbane, Australia, 2015. URL `https://dl.acm.org/citation.cfm?doid=2733373.2806246`.

2014 [6] Nikolaos Pappas, Andrei Popescu-Belis, "Explaining the Stars: Weighted Multiple-Instance Learning for Aspect-Based Sentiment Analysis", In *Proceedings of Conference on Empirical Methods in Natural Language Processing* (EMNLP), pages 455–466, Doha, Qatar, 2014. URL `http://www.aclweb.org/anthology/D14-1052`.

[5] Chidansh A. Bhatt, Nikolaos Pappas, Maryam Habibi and Andrei Popescu-Belis, "Multimodal Reranking of Content-based Recommendations for Hyperlinking Video Snippets", In *Proceedings of ACM International Conference on Multimedia Retrieval* (ICMR), pages 225–232, Glasgow, UK, 2014. URL `https://dl.acm.org/citation.cfm?id=2578752`.

2013 [4] Nikolaos Pappas, Andrei Popescu-Belis, "Sentiment Analysis of User Comments for One-Class Collaborative Filtering over TED Talks", In *Proceedings of 36th ACM SIGIR Conference on Research and Development in Information Retrieval* (SIGIR), pages 773–776, Dublin, Ireland, 2013. URL `https://dl.acm.org/citation.cfm?id=2484116`.

[3] Nikolaos Pappas, Andrei Popescu-Belis, "Combining Content with User Preferences for TED Lecture Recommendation", In *Proceedings of 11th International Workshop on Content Based Multimedia Indexing* (CBMI), pages 47–52, Veszprém, Hungary, 2013. URL `https://ieeexplore.ieee.org/document/6576551/`.

[2] Chidansh A. Bhatt, Andrei Popescu-Belis, Maryam Habibi, Sandy Ingram, Stefano Masneri, Fergus McInnes, Nikolaos Pappas, Oliver Schreer, "Multi-factor Segmentation for Topic Visualization and Recommendation: the MUST-VIS System", In *Proceedings of 21st ACM International Conference on Multimedia* (ACMMM), pages 365–368, Barcelona, Spain, 2013. URL `https://dl.acm.org/citation.cfm?id=2508120`.

[1] Chidansh A. Bhatt, Nikolaos Pappas, Maryam Habibi and Andrei Popescu-Belis, "Idiap at MediaEval 2013: Search and Hyperlinking Task", In *Proceedings of MediaEval 2013 Workshop*, Barcelona (MediaEval), Spain, 2013. URL `http://ceur-ws.org/Vol-1043/mediaeval2013_submission_47.pdf`.

*: equal contribution.

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